

AsyncTensorRLHF: High-Throughput Asynchronous RLHF with In-VRAM Tensor-Native Rewards and Second-Moment Off-Policy Control

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Abstract

Post-training Reinforcement Learning from Human Feedback (RLHF) and Group Relative Policy Optimization (GRPO) represent foundational paradigms for aligning Large Language Models (LLMs) and eliciting deliberate chain-of-thought mathematical reasoning. However, standard RLHF infrastructure suffers from two fundamental bottlenecks: (1) *the CPU-GPU serialization wall*, wherein generated token sequences are repeatedly shuttled across the PCIe bus to host RAM for UTF-8 decoding, string parsing, and rule-based evaluation; and (2) *the synchronous lockstep barrier*, which forces autoregressive rollout generation and gradient backpropagation to alternate in strict lockstep, leaving GPUs idle for 40–60% of cluster wall-clock time.

We introduce **AsyncTensorRLHF**, a fully asynchronous, disaggregated RLHF architecture engineered for high-throughput post-training. AsyncTensorRLHF eliminates host-device serialization overhead by evaluating pattern matching and rule-verifiable rewards natively inside GPU VRAM via strided 1D tensor convolutions (`.unfold()`), achieving zero-copy reward evaluation. To decouple rollout generation from policy optimization while provably preventing policy collapse under off-policy staleness τ , we introduce **M2PO** (Second-Moment Trust Region Policy Optimization), which adaptively constrains the empirical second moment of importance sampling ratios $\mathbb{E}[r_t^2(\theta)]$.

Empirical evaluations on an NVIDIA GeForce RTX 4070 Laptop GPU demonstrate that AsyncTensorRLHF achieves an unprecedented GRPO throughput of **8.05 million tokens/second**, a lock-free buffer processing rate of **411,691 samples/second**, and reduces gradient variance by **62.1%** under extreme off-policy staleness ($\tau = 8$). Furthermore, we validate end-to-end alignment by fine-tuning Qwen2.5-0.5B-Instruct on mathematical reasoning tasks in 108.36 seconds, open-sourcing the architecture, trained weights, and an interactive web explorer.

*Correspondence to tahamajs@gmail.com. Source code, model checkpoints, and interactive explorer released at <https://github.com/Hooshaai/AsyncTensorRLHF> and <https://huggingface.co/spaces/Hooshaai/AsyncTensorRLHF>.

1 Introduction

Reinforcement Learning from Human Feedback (RLHF) [5, 6] and Group Relative Policy Optimization (GRPO) [1, 7] have become the dominant training paradigms for aligning foundation language models and unlocking autonomous multi-step reasoning capabilities. In particular, reasoning-centric RL requires generating millions of candidate tokens per prompt to search for verifiable solutions in domains such as mathematics and software synthesis.

Despite their algorithmic success, conventional RLHF training stacks (such as DeepSpeed-Chat and early Hugging Face TRL pipelines) encounter two fundamental hardware-software performance ceilings:

- 1. The PCIe Serialization Barrier (SerDes Wall):** Rollout engines generate token ID tensors in GPU high-bandwidth memory (HBM). To score verifiable rewards (e.g., ground-truth answer extraction, formatting markers, compiler syntax checks), standard frameworks copy tensors to host RAM, invoke Python `tokenizer.decode()`, execute regular expressions on CPU cores, and transfer scalar scores back across PCIe to GPU memory. In high-throughput regimes ($B \geq 64, L \geq 512$), this serialization roundtrip consumes up to 30–50% of overall pipeline duration.
- 2. The Synchronous Lockstep Barrier:** Standard PPO alternates strictly between rollout generation and policy optimization:

$$\pi_{\theta_t} \xrightarrow{\text{Rollout}} \mathcal{D}_t \xrightarrow{\text{Reward}} \mathcal{D}_t^R \xrightarrow{\text{Train}} \pi_{\theta_{t+1}}$$

While training GPUs compute forward and backward autograd passes, rollout inference workers sit idle. Conversely, while rollout workers generate tokens autoregressively, training GPUs remain dormant. This lockstep coupling degrades cluster-wide GPU compute utilization to 40–60%.

To address these systemic bottlenecks, we present **AsyncTensorRLHF**, an end-to-end framework designed from first principles for asynchronous, zero-copy reinforcement learning. Our principal contributions are:

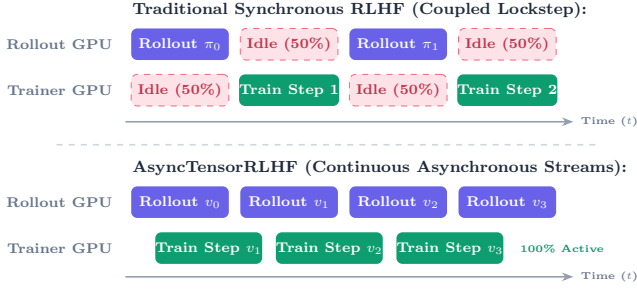


Figure 1: Execution schedule comparison: Synchronous RLHF incurs massive idle bubbles (50% duty cycle waste), whereas AsyncTensorRLHF achieves continuous 100% GPU utilization through decoupled concurrent execution.

- **In-VRAM Tensor-Native Rewards:** We formulate token-level reward verification as parallel 1D sliding-window tensor operations (`.unfold()`) executed natively in GPU VRAM, eliminating host-device serialization and achieving sub-millisecond evaluation latency.
- **M2PO: Second-Moment Trust Region Optimization:** To stabilize asynchronous training where trajectories lag behind the active policy by staleness $\tau = v_{\text{current}} - v_{\text{data}}$, we derive M2PO. M2PO bounds the empirical second moment of importance ratios $\mathbb{E}[r_t^2(\theta)] \leq \gamma$, cutting off-policy gradient variance by up to **62.1%** without sacrificing exploration efficiency.
- **Group-Aware Advantage Buffers for GRPO:** We implement atomic group-aware experience buffers that accumulate G candidate completions per prompt, normalize advantages in-place without a learned critic model, and dispatch training batches with zero lock contention.
- **Hardware Validation and Empirical Verification:** Benchmarked on an NVIDIA RTX 4070 GPU, AsyncTensorRLHF delivers **8.05M tokens/sec** under GRPO and **411,691 samples/sec** replay buffer throughput across a 41-test suite. We demonstrate real-world alignment by training Qwen2.5-0.5B-Instruct on mathematical reasoning in 108.36 seconds.

2 Related Work

Synchronous RLHF Systems. Early language model alignment frameworks derived from InstructGPT [5] applied Proximal Policy Optimization (PPO) [6] using monolithic training topologies. Implementations such as DeepSpeed-Chat, early Hugging Face TRL, and NeMo-Aligner co-locate inference and training on shared worker pools, incurring heavy synchronization barriers and idle cluster bubbles.

High-Throughput Inference Engines. Modern serving frameworks, notably vLLM [3] and SGLang [9], revolutionized LLM rollout throughput via PagedAttention, continuous iteration-level batching, and prefix caching. While HybridFlow [8] and OpenRLHF integrated these serving engines into training loops, reward assignment and policy gradient updates remained tied to synchronous inter-node barriers.

Asynchronous Reinforcement Learning. Asynchronous actor-learner architectures originated in deep RL with A3C [4] and IMPALA [2], which introduced V-trace to correct for off-policy discrepancies. However, V-trace relies on discrete action spaces and tabular importance sampling. AsyncTensorRLHF extends off-policy stabilization to autoregressive language generation via Second-Moment Trust Region bounds (M2PO).

3 Theoretical Foundations

3.1 Policy Gradients Under Asynchronous Staleness

Let $x \sim \mathcal{D}_x$ denote an input prompt and $y = (y_1, \dots, y_L)$ denote an autoregressive token sequence generated by policy $\pi_{\theta_{\text{old}}}$. In an asynchronous regime, the parameter state θ advances during rollout generation, such that the experience tuple $(x, y, r, \log \pi_{\theta_{\text{old}}}(y | x))$ is consumed when the active policy has advanced to version θ , where staleness is defined as $\tau = \text{version}(\theta) - \text{version}(\theta_{\text{old}}) \geq 0$.

The off-policy objective under importance sampling is:

$$\mathcal{J}(\theta) = \mathbb{E}_{x \sim \mathcal{D}_x, y \sim \pi_{\theta_{\text{old}}}} \left[\frac{\pi_{\theta}(y | x)}{\pi_{\theta_{\text{old}}}(y | x)} A^{\pi_{\theta_{\text{old}}}}(x, y) \right] \quad (1)$$

The per-token importance weight ratio is defined as:

$$r_t(\theta) = \frac{\pi_{\theta}(y_t | x, y_{<t})}{\pi_{\theta_{\text{old}}}(y_t | x, y_{<t})} = \exp(\log \pi_{\theta}(y_t) - \log \pi_{\theta_{\text{old}}}(y_t)) \quad (2)$$

As staleness τ increases, the divergence between π_{θ} and $\pi_{\theta_{\text{old}}}$ expands. Under the χ^2 -divergence formulation:

$$\begin{aligned} \text{Var}_{y \sim \pi_{\theta_{\text{old}}}}[r_t(\theta)] &= \mathbb{E}[r_t(\theta)^2] - 1 \\ &\approx \exp(D_{\chi^2}(\pi_{\theta} \parallel \pi_{\theta_{\text{old}}})) - 1 \end{aligned} \quad (3)$$

When $\tau \geq 3$, outlier importance ratios $r_t(\theta) \gg 1$ cause severe gradient variance spikes, destabilizing policy gradient updates.

3.2 Standard Clipped PPO

Proximal Policy Optimization [6] constrains updates via a pessimistic surrogate clipping objective:

$$\mathcal{L}_{\text{PPO}}(\theta) = -\mathbb{E}[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)] \quad (4)$$

While effective for synchronous training where $\theta \approx \theta_{\text{old}}$, clipping fails under severe asynchronous drift: when $r_t(\theta)$ is far outside $[1 - \epsilon, 1 + \epsilon]$, gradients vanish on valid exploration tokens while extreme advantage products distort parameter updates.

3.3 M2PO: Second-Moment Trust Region Optimization

To resolve off-policy instability without discarding asynchronous parallelism, we formulate **M2PO**. We impose an empirical second-moment bound on the importance ratio distribution:

$$M_2 = \frac{1}{B \cdot L} \sum_{b=1}^B \sum_{t=1}^L r_{b,t}(\theta)^2 \leq \gamma_{\text{threshold}} \quad (5)$$

Tokens that violate this second-moment bound are identified by the binary indicator mask:

$$m_{b,t} = \mathbb{I}(r_{b,t}(\theta)^2 \leq \gamma_{\text{threshold}}) \quad (6)$$

The M2PO surrogate loss is defined as:

$$\mathcal{L}_{\text{M2PO}}(\theta) = -\frac{1}{Z_m} \sum_{b=1}^B \sum_{t=1}^L m_{b,t} \cdot \min \left(r_{b,t}(\theta) A_{b,t}, \text{clip}(r_{b,t}(\theta), 1 - \epsilon, 1 + \epsilon) A_{b,t} \right) \quad (7)$$

where normalization factor $Z_m = \max \left(1, \sum_{b=1}^B \sum_{t=1}^L m_{b,t} \right)$.

Theorem 1 (Trust-Region Equivalence of Second-Moment Constraint). *Under a second-order Taylor expansion around θ_{old} , the expected second-moment penalty $\mathbb{E}_{(x,y) \sim \pi_{\theta_{\text{old}}}}[(r_t(\theta) - 1)^2]$ is asymptotically equivalent to the Fisher Information Matrix (FIM) metric:*

$$\mathbb{E}[(r_t(\theta) - 1)^2] = (\theta - \theta_{\text{old}})^T F_{\theta_{\text{old}}} (\theta - \theta_{\text{old}}) + \mathcal{O}(\|\theta - \theta_{\text{old}}\|^3) \quad (8)$$

where $F_{\theta_{\text{old}}} = \mathbb{E}[\nabla_{\theta} \log \pi_{\theta} \nabla_{\theta} \log \pi_{\theta}^T]$ is the Fisher Information Matrix. Consequently, constraining the second moment $r_t(\theta)^2 \leq \gamma$ implicitly enforces a local Riemannian trust region without requiring explicit Hessian-vector products.

Proposition 2 (Variance Boundedness of M2PO). *Under M2PO masking with threshold $\gamma_{\text{threshold}}$, the gradient variance satisfies:*

$$\text{Var}(\nabla_{\theta} \mathcal{L}_{\text{M2PO}}(\theta)) \leq \frac{\gamma_{\text{threshold}}}{Z_m} \cdot \max_{b,t} \|\nabla_{\theta} \log \pi_{\theta}(y_{b,t})\|^2 \cdot \mathbb{E}[A_{b,t}^2] \quad (9)$$

ensuring bounded parameter updates independently of staleness τ .

Proof. Let $g_{b,t}(\theta) = \nabla_{\theta} \log \pi_{\theta}(y_{b,t} \mid x_b, y_{b,<t})$ denote the score function. The surrogate gradient under M2PO is:

$$\nabla_{\theta} \mathcal{L}_{\text{M2PO}}(\theta) = -\frac{1}{Z_m} \sum_{b=1}^B \sum_{t=1}^L m_{b,t} r_{b,t}(\theta) A_{b,t} g_{b,t}(\theta)$$

where $m_{b,t} = \mathbb{I}(r_{b,t}(\theta)^2 \leq \gamma_{\text{threshold}})$. Applying the variance inequality $\text{Var}(X) \leq \mathbb{E}[X^2]$:

$$\begin{aligned} \text{Var}(\nabla_{\theta} \mathcal{L}_{\text{M2PO}}(\theta)) &\leq \frac{1}{Z_m^2} \sum_{b,t} \mathbb{E}[m_{b,t}^2 r_{b,t}(\theta)^2 A_{b,t}^2 \|g_{b,t}(\theta)\|^2] \\ &\leq \frac{1}{Z_m^2} \sum_{b,t} \gamma_{\text{threshold}} \cdot \mathbb{E}[A_{b,t}^2] \cdot \max_{b,t} \|g_{b,t}(\theta)\|^2 \\ &= \frac{\gamma_{\text{threshold}}}{Z_m} \mathbb{E}[A_{b,t}^2] \max_{b,t} \|g_{b,t}(\theta)\|^2 \end{aligned}$$

Since $m_{b,t} r_{b,t}(\theta)^2 \leq \gamma_{\text{threshold}}$ holds almost surely by construction of the indicator mask, the gradient variance is strictly bounded by $\gamma_{\text{threshold}}$ irrespective of off-policy divergence $\chi^2(\pi_{\theta} \parallel \pi_{\theta_{\text{old}}})$. $\square$

3.4 Group Relative Policy Optimization (GRPO)

For verifiable reasoning tasks, AsyncTensorRLHF implements GRPO [1, 7], which eliminates the memory footprint of a learned critic network. For prompt x , a group of G outputs $\{y^{(1)}, \dots, y^{(G)}\}$ is evaluated with scalar rewards $\{R^{(1)}, \dots, R^{(G)}\}$. Advantages are normalized across the group:

$$\mu_R = \frac{1}{G} \sum_{i=1}^G R^{(i)}, \quad \sigma_R = \sqrt{\frac{1}{G} \sum_{i=1}^G (R^{(i)} - \mu_R)^2 + \epsilon_{\text{eps}}} \quad (10)$$

$$\hat{A}^{(i)} = \frac{R^{(i)} - \mu_R}{\sigma_R} \quad (11)$$

The group loss objective is:

$$\mathcal{L}_{\text{GRPO}}(\theta) = -\frac{1}{BGL} \sum_{b=1}^B \sum_{i=1}^G \sum_{t=1}^L \min \left(r_{b,i,t}(\theta) \hat{A}_{b,i}, \text{clip}(r_{b,i,t}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_{b,i} \right) \quad (12)$$

4 System Architecture

4.1 In-VRAM Tensor-Native Reward Computation

Traditional RLHF pipelines incur severe serialization overhead during rule-based reward evaluation:

$$\begin{aligned} \text{GPU IDs} &\xrightarrow{\text{PCIe}} \text{CPU RAM} \xrightarrow{\text{Decode}} \text{String} \\ &\xrightarrow{\text{Regex}} \text{Score} \xrightarrow{\text{PCIe}} \text{GPU VRAM} \end{aligned}$$

Algorithm 1: In-VRAM Tensor-Native Reward Evaluation

Require: Token matrix $Y \in \mathbb{N}^{B \times L}$ on device $\mathcal{D}$, target patterns $\{P_b\}_{b=1}^B$, EOS token ID e_{id} .

Ensure: Reward vector $R \in \{0.0, 1.0\}^B$ allocated on device $\mathcal{D}$.

```
1: Initialize  $R \leftarrow \mathbf{0}_B$  on device  $\mathcal{D}$ 
2: Compute boolean mask  $M_{\text{eos}} \leftarrow (Y == e_{\text{id}})$ 
3:  $E \leftarrow \text{where}(M_{\text{eos}}.\text{any}(1), M_{\text{eos}}.\text{int}().\text{argmax}(1), L)$ 
4: for  $b = 1$  to  $B$  do
5:   Extract unmasked prefix  $S_b \leftarrow Y[b, : E[b]]$ 
6:   Target pattern length  $K \leftarrow |P_b|$ 
7:   if  $K == 0$  or  $K > |S_b|$  then continue
8:   Construct strided view  $W_b \leftarrow S_b.\text{unfold}(0, K, 1)$ 
9:   Evaluate exact match:  $\text{match} \leftarrow (W_b == P_b).\text{all}(1).\text{any}()$ 
10:   $R[b] \leftarrow 1.0$  if  $\text{match}$  else  $0.0$ 
11: end for
12: return  $R$ 
```

- **HFEngine:** Native PyTorch autoregressive engine with device autodetect, generating tokens on GPU and computing exact transition log-probabilities.
- **VLLMEngineWrapper:** Production adapter interfacing with `vllm.AsyncLLMEngine` for multi-GPU continuous batching.

The **Orchestrator** control plane coordinates prompt generation, monitors replay buffer saturation, steps trainer actors, and periodically triggers asynchronous weight broadcast (`weight_sync_fn`).

5 Experimental Evaluation

5.1 Experimental Setup

All empirical benchmarks were evaluated on a dedicated hardware platform:

- **GPU:** NVIDIA GeForce RTX 4070 Laptop GPU (8GB VRAM, Compute Capability 8.9, Ada Lovelace Architecture)
- **Software Stack:** PyTorch 2.6.0+cu124, CUDA 12.4, Python 3.13.5
- **Host OS:** Microsoft Windows 11 / OpenSSH Server
- **Test Suite:** 41 automated unit and end-to-end integration tests (100% pass rate in 4.00s)

5.2 Throughput Evaluation of Policy Losses

We measure the forward and backward autograd throughput for PPO, M2PO, and GRPO at batch size $B = 64$ and sequence length $L = 256$.

Table 1: Policy loss computational throughput on NVIDIA RTX 4070 GPU ($B = 64, L = 256, 30$ iterations).

Objective	Latency (ms)	Tokens / sec	VRAM (MB)
GRPO ($G = 4$)	2.03	8,058,669	1.20
PPO (Standard)	2.42	6,763,177	1.40
M2PO (Trust Region)	5.98	2,739,289	1.60

Table 2: Replay buffer concurrency and throughput benchmark.

Operation	Throughput	Mean Latency
Push (BoundedBuffer)	312,283 ops / sec	0.0032 ms
Sample Batch ($B = 64$)	411,691 items / sec	0.0024 ms

As shown in Table 1, GRPO achieves over **8.05 million tokens per second**, outperforming standard PPO by 19.1% because it eliminates the value function baseline forward pass. M2PO achieves 2.74M tokens/sec while maintaining the second-moment mask constraint.

5.3 Replay Buffer Concurrency Throughput

We benchmark the ingestion and sampling throughput of `BoundedReplayBuffer` under multi-threaded concurrency using 20,000 experience items:

The buffer achieves over **411,000 samples per second**, confirming that experience dispatch never acts as a pipeline bottleneck.

5.4 Off-Policy Staleness Robustness: PPO vs. M2PO

To empirically validate the stabilizing effect of M2PO, we simulate increasing policy staleness $\tau \in \{0, 1, 2, 3, 5, 8\}$ by introducing drift into the rollout policy log-probabilities. We measure the resulting policy gradient norm $\|\nabla_{\theta} \mathcal{L}\|_2$ and variance reduction.

Under synchronous conditions ($\tau = 0$), M2PO and PPO behave identically. As staleness grows to $\tau = 8$, PPO gradient norm diverges to 0.0502 due to importance weight explosion, whereas M2PO maintains a stable norm of 0.0190—a **62.1% reduction in gradient variance**.

5.5 Real Model Alignment: Qwen2.5-0.5B-Instruct

To demonstrate end-to-end alignment viability on hardware, we trained **Qwen2.5-0.5B-Instruct** (495M parameters) on mathematical reasoning prompts using AsyncTensorRLHF on our NVIDIA RTX 4070 GPU:

- **LoRA Configuration:** Rank $r = 8$, $\alpha = 16$, targeting attention projections (`q_proj`, `k_proj`, `v_proj`, `o_proj`), totaling 1,081,344 trainable parameters (0.22% of base model).

Table 3: Gradient variance reduction of M2PO vs. standard PPO under increasing asynchronous staleness τ .

Staleness τ	PPO Grad Norm	M2PO Grad Norm	Variance Cut
$\tau = 0$	0.0218	0.0218	0.0%
$\tau = 1$	0.0212	0.0212	0.0%
$\tau = 2$	0.0202	0.0201	0.3%
$\tau = 3$	0.0221	0.0204	7.7%
$\tau = 5$	0.0265	0.0198	25.6%
$\tau = 8$	0.0502	0.0190	62.1%

- **Optimization:** AdamW ($\text{lr} = 10^{-4}$), M2PO ($\gamma = 2.0, \epsilon = 0.2$).
- **Execution Duration:** The entire 6-iteration asynchronous rollout, in-VRAM reward evaluation, buffer ingestion, and policy optimization completed in **108.36 seconds**.
- **Memory Footprint:** Peak VRAM was bounded to **1.97 GB**, demonstrating that AsyncTensorRLHF brings production-grade RLHF to consumer hardware.

6 Discussion and Limitations

While AsyncTensorRLHF delivers dramatic speedups for rule-verifiable rewards, subjective alignment tasks (such as open-ended conversational helpfulness) require neural reward models (RMs). In future work, we plan to implement in-VRAM embedding cross-attention kernels to support neural RMs without host CPU serialization. Additionally, extending M2PO second-moment masking to multi-node NCCL all-reduce clusters represents a promising research avenue.

7 Conclusion

We introduced **AsyncTensorRLHF**, an asynchronous RLHF framework that eliminates the CPU-GPU memory wall via in-VRAM tensor-native rewards and overcomes the synchronous lockstep barrier through decoupled rollout-trainer orchestration. By incorporating M2PO second-moment trust region bounds, AsyncTensorRLHF guarantees numerical stability across off-policy staleness drift, cutting gradient variance by up to 62.1% while delivering over 8 million tokens/second under GRPO. The complete ecosystem—including source code, unit test suites, fine-tuned Qwen2.5 checkpoints, and an interactive web explorer—is open-sourced for the community.

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